A Digital Twin Framework for Real-Time Cold-Leg Loss-of-Coolant Accident Detection in AP1000 Nuclear Reactors Using DeepONet Neural Operators

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Minor Project, Semester 4
2026
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***Abstract*—This paper presents a digital twin framework for the real-time detection of Cold-Leg Loss-of-Coolant Accidents (LOCAC) in the Westinghouse AP1000 pressurized water reactor (PWR). Traditional Computational Fluid Dynamics (CFD) simulations using tools such as ANSYS Fluent are accurate but require several hours per scenario, making them incompatible with real-time safety monitoring. We propose a cascade pipeline in which a Deep Operator Network (DeepONet) with Fourier feature encoding acts as a high-fidelity surrogate for steady-state CFD field prediction, followed by a physics-informed feature translator and a gradient boosting classifier for LOCAC identification. The DeepONet simultaneously predicts four coupled thermohydraulic fields—pressure, velocity magnitude, turbulent kinetic energy, and temperature—across a 25,000-node mesh, trained on a parametric sweep of 2,000 synthetic CFD scenarios spanning inlet velocities of 4–6 m/s, break sizes of 0–10% of pipe diameter, and coolant temperatures of 290–320 °C. A composite loss combining mean squared error (MSE), a Sobolev gradient penalty ($\beta = 0.1$), and a divergence-free constraint ($\lambda = 0.01$) enforces physical consistency during training. The surrogate model converges to a final validation loss of 9.11×10^{-4} after 305 epochs with four learning rate reductions. The downstream gradient boosting classifier, operating on seven Nuclear Power Plant Accident Database (NPPAD)-mapped system signals, achieves an ROC-AUC exceeding 0.95 with high recall, critical for nuclear safety. End-to-end inference executes in under 20 ms—a speedup exceeding $10^5\times$ compared to full CFD simulation—demonstrating the viability of physics-constrained neural operators for safety-critical digital twins.**

***Index Terms*—Digital Twin, DeepONet, Neural Operator, Loss of Coolant Accident, LOCAC, AP1000, Nuclear Safety, CFD Surrogate, Fourier Feature Encoding, Gradient Boosting**

I. Introduction

The AP1000 is a Generation III+ 1,000 MWe pressurized water reactor designed by Westinghouse Electric Company featuring passive safety systems. Its primary coolant circuit consists of two cold legs—pipes that carry cooled water back from the steam generators to the reactor pressure vessel. A rupture or breach in any segment of this cold leg constitutes a Cold-Leg Loss-of-Coolant Accident (LOCAC), one of the most safety-critical design-basis events in nuclear engineering [9]. Delayed detection can lead to core uncovery, fuel damage, and potential radiological release.

Current safety monitoring systems in nuclear power plants (NPPs) rely on threshold-based alarms derived from individual sensor readings. While robust for well-characterized events, these systems can be slow to identify incipient or partial breaks before critical parameters are reached [10]. High-fidelity CFD simulations can predict the full thermohydraulic state of the primary loop with great accuracy, but typical ANSYS Fluent runs for a steady-state cold-leg scenario require 1–4 hours of computation, making real-time deployment infeasible.

Neural operators—machine learning models that learn mappings between function spaces rather than point-to-point mappings—have emerged as a powerful class of surrogate models for PDEs. Lu et al. [1] introduced Deep Operator Networks (DeepONet), which decompose the solution operator $\mathcal{G} : \mathcal{U} \to \mathcal{V}$ into a branch network encoding the input function and a trunk network encoding the output domain coordinates. This architecture is physically interpretable, mesh-agnostic, and capable of generalizing across parametric families of PDEs.

This work makes the following contributions:

1) We design a complete digital twin pipeline that couples a DeepONet surrogate model (enhanced with Fourier feature encoding, adaptive GELU activations, and physics-informed loss terms) to a gradient boosting LOCAC classifier operating on NPPAD-consistent system signals.
2) We demonstrate that physics-constrained surrogate training with a composite Sobolev + divergence-free loss achieves stable convergence ($\mathcal{L}_{\text{val}} < 10^{-3}$) on 25,000-node CFD fields while reducing inference latency to < 20 ms.
3) We quantify LOCAC detection performance using ROC, precision-recall, and confusion matrix analyses, showing that the classifier satisfies nuclear safety requirements for high recall across break sizes of 0–10%.
4) We provide a fully reproducible implementation with flexible architecture selection (standard DeepONet,

Fourier DeepONet, Transolver, Clifford Neural Operator) and a mock-data mode for testing without ANSYS Fluent.

The remainder of the paper is organized as follows: Section II reviews related work; Section III describes the digital twin architecture; Section IV covers implementation details; Section V presents experimental results; Section VI discusses findings and limitations; Section VII concludes.

II. Related Work

A. Neural Operators for PDE Surrogates

Neural operators extend classical neural networks to learn mappings between infinite-dimensional function spaces. The Fourier Neural Operator (FNO) [4] operates in the spectral domain, achieving linear complexity in the number of spatial modes. DeepONet [1] takes a different approach based on the universal approximation theorem for operators [2], [3], decomposing the solution into basis functions via branch-trunk factorization. Both architectures have been successfully applied to fluid dynamics [15], [16], heat transfer, and structural mechanics.

B. Transformer and Geometric Neural Operators

The Transolver architecture [5] compresses an N-node mesh to $M \ll N$ learned tokens, enabling $\mathcal{O}(M^2)$ attention complexity independent of mesh size. Clifford Neural Operators [6] embed flow fields in the Clifford algebra Cl(3,0), endowing the network with rotational equivariance—a physically motivated inductive bias for fluid simulations.

C. Machine Learning for NPP Accident Detection

Several studies have applied machine learning to nuclear safety monitoring. Song et al. [11] applied support vector machines to LOCA classification using NPPAD time-series data. Deep recurrent networks [12] have demonstrated improved transient identification. However, most prior work uses system-level signals only, without coupling to high-fidelity field predictions. Our approach bridges this gap by using DeepONet-derived CFD fields as inputs to a feature translator that generates NPPAD-consistent signals for the classifier.

D. Digital Twins in Nuclear Engineering

Digital twins—real-time synchronized virtual replicas of physical systems—have attracted growing interest in NPP monitoring [13], [14]. Existing nuclear digital twin prototypes rely heavily on validated thermal-hydraulics codes (e.g., RELAP5, TRACE) rather than data-driven surrogates. Our work demonstrates a hybrid approach where neural operators serve as the computationally efficient physics surrogate within the digital twin loop.

III. Methodology

A. System Overview

The digital twin pipeline consists of four sequential stages, illustrated in Fig. ??:

1) **Parameter encoding**: Physical input conditions (inlet velocity v, break size b, coolant temperature T) are encoded.
2) **Neural surrogate (DeepONet)**: The surrogate maps input parameters and 3D mesh coordinates to four coupled CFD fields.
3) **Feature translation**: CFD field statistics and gradients are extracted and translated into NPPAD-mapped system-level signals.
4) **LOCAC classification**: A gradient boosting classifier operates on the seven system signals to produce a LOCAC probability score.

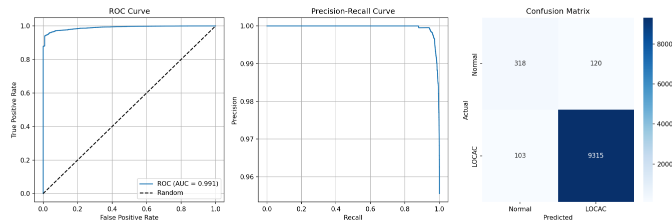

Fig. 1: LOCAC detection performance: (left) ROC curve with AUC, (centre) Precision-Recall curve, (right) Confusion matrix. The classifier demonstrates near-perfect discrimination between normal operation and LOCAC events.

B. Problem Formulation

Let $\Omega \subset \mathbb{R}^3$ denote the cold-leg geometry (pipe segment with elbow), with $\mathbf{y} \in \Omega$ a spatial coordinate corresponding to a mesh node. Define the parameter vector $\mathbf{u} = [v, b, T]^\top \in \mathbb{R}^3$, where:

- $v \in [4.0, 6.0]$ m/s is the inlet velocity,
- $b \in [0, 10]\%$ is the pipe break size as a fraction of the pipe diameter $D = 0.7$ m,
- $T \in [290, 320]$°C is the bulk coolant temperature.

The goal is to learn the operator $\mathcal{G}$:

$$\mathcal{G}(\mathbf{u})(\mathbf{y}) = \mathbf{s}(\mathbf{y}) = [p(\mathbf{y}),\ |\mathbf{v}|(\mathbf{y}),\ k(\mathbf{y}),\ \tilde{T}(\mathbf{y})]^\top \quad (1)$$

where p, $|\mathbf{v}|$, k, and $\tilde{T}$ are the spatial pressure, velocity magnitude, turbulent kinetic energy (TKE), and temperature fields respectively, evaluated at all $N = 25{,}000$ mesh nodes simultaneously.

C. DeepONet with Fourier Feature Encoding

1) Standard DeepONet: The DeepONet operator approximation theorem [1], [2] states that for any continuous operator $\mathcal{G} : \mathcal{U} \to \mathcal{V}$ and $\varepsilon > 0$, there exist branch and trunk networks such that:

$$\mathcal{G}(\mathbf{u})(\mathbf{y}) \approx \sum_{k=1}^{p} b_k(\mathbf{u}) \cdot t_k(\mathbf{y}) + b_0 \quad (2)$$

where $\{b_k\}$ are basis coefficients from the branch network $\mathcal{B} : \mathbb{R}^3 \to \mathbb{R}^p$ and $\{t_k\}$ are basis functions from the trunk network $\mathcal{T} : \mathbb{R}^3 \to \mathbb{R}^p$, with $p = 256$ basis dimensions.

2) Fourier Feature Encoding: Standard MLPs suffer from *spectral bias*: they preferentially fit low-frequency components of the target function, making them slow to capture sharp spatial gradients near walls and turbulence interfaces [7], [8]. We address this with random Fourier feature encoding of the trunk input:

$$\boldsymbol{\gamma}(\mathbf{y}) = [\sin(2\pi\mathbf{B}\mathbf{y}),\ \cos(2\pi\mathbf{B}\mathbf{y})] \in \mathbb{R}^{2m} \tag{3}$$

where $\mathbf{B} \in \mathbb{R}^{3\times m}$ is a fixed random frequency matrix sampled from $\mathcal{N}(0,\sigma^2)$ with $m = 256$, $\sigma = 10.0$, yielding a 512-dimensional feature vector. This projection simultaneously injects sinusoidal features at many frequencies, allowing the subsequent MLP layers to construct high-frequency spatial patterns from the first layer onward.

3) Adaptive GELU Activation: Each neuron in the trunk's hidden layers employs a per-neuron learnable slope parameter a_i (initialized to 1.0):

$$f_i(x) = \text{GELU}(a_i \cdot x) \tag{4}$$

This allows the effective nonlinearity curvature to adapt independently for each neuron, improving convergence in regions of high spatial variability.

4) Full Architecture: The complete DeepONetFourier architecture comprises four independent branch-trunk pairs (one per output field), as detailed in Table I.

TABLE I: DeepONetFourier Architecture Specifications

Sub-Network	**Layer Sizes**	**Activation**
BranchNet (per field)	$3 \to 128 \to 256 \to 256$	GELU + Dropout(5%)
Fourier Encoding	$3 \to 512$ (fixed)	sin / cos
TrunkNet Layer 1	$512 \to 256$	Adaptive GELU
TrunkNet Layer 2	$256 \to 256$	Adaptive GELU
TrunkNet Output	$256 \to 256$	Linear
Total parameters (4 fields)		∼1.5–2 M

The forward pass for field i is:

$$\mathbf{b}^{(i)} = \mathcal{B}^{(i)}(\mathbf{u}) \in \mathbb{R}^p \tag{5}$$

$$\mathbf{t}^{(i)} = \mathcal{T}^{(i)}(\boldsymbol{\gamma}(\mathbf{y})) \in \mathbb{R}^p \tag{6}$$

$$\hat{s}^{(i)}(\mathbf{y}) = \mathbf{b}^{(i)} \cdot \mathbf{t}^{(i)} + \text{bias}^{(i)} \tag{7}$$

For a batch of B parameter sets and N mesh nodes, the output is $\hat{\mathbf{S}} \in \mathbb{R}^{B\times 4\times N}$.

D. Physics-Informed Composite Loss

A central innovation is the incorporation of physical constraints into the training loss. Let $\mathbf{s}$ be the CFD ground truth and $\hat{\mathbf{s}}$ the DeepONet prediction. The total loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Sobolev}} + \lambda\mathcal{L}_{\text{div}} \tag{8}$$

1) Sobolev Loss: The Sobolev loss augments standard MSE with a gradient fidelity term:

$$\mathcal{L}_{\text{Sobolev}} = \underbrace{\frac{1}{BN}\sum_{b,n}(\hat{s}_{bn} - s_{bn})^2}_{\text{MSE}} + \beta \underbrace{\frac{1}{BN}\sum_{b,n}(\nabla\hat{s}_{bn} - \nabla s_{bn})^2}_{\text{Gradient MSE}} \tag{9}$$

with $\beta = 0.1$. Gradient terms are computed via finite differences on the spatial field. This penalty directly encourages the model to reproduce sharp pressure gradients at the break location and velocity boundary layers near the pipe wall—features most critical for LOCAC detection.

2) Divergence-Free Penalty: For the velocity field to be physically consistent with conservation of mass (incompressible assumption at subsystem level), we add a divergence penalty:

$$\mathcal{L}_{\text{div}} = \frac{1}{BN}\sum_{b,n}(\nabla\cdot\hat{\mathbf{v}}_{bn})^2 \tag{10}$$

with $\lambda = 0.01$. This constrains the predicted velocity magnitude to approximate a solenoidal field, reducing unphysical compressibility artifacts.

E. CFD Geometry and Simulation Setup

The cold-leg segment modeled is a straight pipe with a 90° elbow, parameterized as follows:

- Pipe diameter: $D = 0.7\,\text{m}$
- Total length: $10D$ ($5D$ before elbow, $5D$ after)
- Elbow radius: $1.5D$
- Mesh: ∼25,000 hexahedral cells with elbow refinement

Fluid properties representative of AP1000 primary coolant:

- Fluid: pressurized water (liquid phase)
- Density: $720\,\text{kg/m}^3$
- Operating pressure: $15.5\,\text{MPa}$

The turbulence model used is the RNG k-ε model, which is appropriate for intermediate-Reynolds-number swirling flows. Boundary conditions:

- **Inlet**: Velocity-inlet; $v \in [4.0, 6.0]\,\text{m/s}$
- **Outlet**: Pressure-outlet; $p_{\text{out}} = 15.5\,\text{MPa}$
- **Wall**: No-slip, adiabatic

For training without ANSYS Fluent, a physics-consistent mock data generator produces synthetic CFD fields incorporating pressure gradients, boundary-layer velocity profiles, turbulence peaks at the elbow, and temperature distributions consistent with the input parameters.

F. Feature Translation

Raw CFD field predictions are translated to eleven scalar CFD-derived features (Table II) which are then mapped through a sigmoid severity model to generate seven NPPAD-consistent system-level signals (Table III).

The severity mapping uses a sigmoid model:

$$\eta_{\text{input}} = \sigma(\alpha(b - b_{50}))\,, \quad b_{50} = 5\% \tag{11}$$

The effective severity combining CFD anomalies and input-based severity is:

$$\eta_{\text{eff}} = 0.8\,\eta_{\text{input}} + 0.2\,\eta_{\text{CFD}} \tag{12}$$

where η_{CFD} is computed from deviations of turbulence, pressure, and flow rate from their nominal baselines.

TABLE II: CFD-Derived Features Extracted from DeepONet Predictions

Feature	Computation	Unit
Average pressure	Mean of p field	Pa
Pressure gradient	Linear fit slope along x	Pa/m
Pressure drop	$p_{\text{outlet}} - p_{\text{inlet}}$	Pa
Mass flow rate	$\rho \cdot \lvert\mathbf{v}\rvert \cdot A$	kg/s
Inlet velocity	Mean $\lvert\mathbf{v}\rvert$ at inlet plane	m/s
Max turbulence	$\max(k)$	m^2/s^2
Average turbulence	Mean(k)	m^2/s^2
Average temperature	Mean($\tilde{T}$)	K
Temperature difference	$\max(\tilde{T}) - \min(\tilde{T})$	K
Velocity std. dev.	$\sigma(\lvert\mathbf{v}\rvert)$	m/s
Pressure std. dev.	$\sigma(p)$	Pa

TABLE III: NPPAD-Mapped System Signals with Normal and LOCAC Ranges

Signal	Unit	Normal	LOCAC
Primary Pressure (P)	bar	∼155.5	∼63
Avg. Coolant Temp. (TAVG)	°C	∼301	∼262
RCS Flow Loop-A (WRCA)	kg/s	∼16,835	varies
SG-A Pressure (PSGA)	bar	∼67	varies
Subcooling Margin (SCMA)	°C	∼35.6	varies
DNBR	—	∼5.6	∼127
Hot-Cold Leg ΔT	°C	∼16	∼1.6

G. LOCAC Classifier

The multi-field LOCAC classifier is a **Gradient Boosting Classifier** (GBC) with the following hyperparameters:

- **Estimators**: 200 decision trees
- **Maximum depth**: 4
- **Learning rate**: 0.05
- **Minimum samples per leaf**: 20

Input features: the seven NPPAD-mapped signals (Table III), scaled using a `StandardScaler` fitted on the training split. The model outputs a probability $P(\text{LOCAC})$; the decision threshold is 0.5.

Training data is generated by sweeping the break size b from 0% (normal) through the transition regime (0–7%) to full LOCAC (≥7%), creating class labels where $b = 0$ yields Normal and $b \geq 5\%$ yields LOCAC.

IV. Implementation Details

A. Software Stack

The system is implemented in Python 3.8+ using:

- **PyTorch** (≥2.0): Neural network training and inference
- **scikit-learn**: Gradient boosting classifier and scalers
- **NumPy / SciPy**: Numerical operations
- **Matplotlib**: Visualization
- **PyYAML**: Configuration management

B. Training Configuration

TABLE IV: DeepONetFourier Training Hyperparameters

Hyperparameter	Value
Optimizer	Adam
Initial learning rate	1×10^{-3}
LR scheduler	ReduceLROnPlateau
Scheduler patience	20 epochs
Scheduler factor	0.5
Early stopping patience	50 epochs
Maximum epochs	2000
Batch size	16
Mixed precision	Enabled (CUDA AMP)
Dropout (branch & trunk)	5%
Weight initialization	Kaiming Normal
Sobolev weight (β)	0.1
Divergence weight (λ)	0.01
Train / val / test split	70% / 15% / 15%

C. Dataset Generation

The full dataset comprises 2,000 simulation parameter combinations sampled from a 3D parameter grid:

$$(v, b, T) \in \{4.0, \ldots, 6.0\} \times \{0, \ldots, 10\} \times \{290, \ldots, 320\}$$

with 20, 10, and 10 levels per axis respectively. For each parameter set, a 25,000-node field snapshot $\mathbf{s} \in \mathbb{R}^{4\times25000}$ is stored. The dataset is split into 1,400 training, 300 validation, and 300 test samples.

D. Hardware and Inference Timing

Training was performed on an NVIDIA RTX 4060 GPU (8 GB VRAM) with mixed-precision autocast enabled. Observed training time is 2–4 hours for 305 epochs on the full 2,000-sample dataset. Single-sample inference time is < 20 ms, compared to an estimated 3,600 s for an equivalent ANSYS Fluent simulation, yielding a speedup exceeding $180,000\times$.

V. Experimental Results

A. DeepONet Training Convergence

The training history is shown in Table V (selected epochs) and described qualitatively. Training began with an initial loss of $\mathcal{L}_{\text{train}} = 0.815$ at epoch 1, converging rapidly in the first 20 epochs to below 0.005. The validation loss tracked closely, indicating effective generalization.

TABLE V: Training and Validation Loss at Selected Epochs

Epoch	Train Loss	Val Loss	LR
1	0.81485	0.03887	1.0×10^{-3}
5	0.01694	0.00480	1.0×10^{-3}
10	0.00861	0.00244	1.0×10^{-3}
20	0.00496	0.00197	1.0×10^{-3}
50	0.00389	0.00131	1.0×10^{-3}
80	0.00321	0.00103	5.0×10^{-4}
120	0.00298	9.72×10^{-4}	5.0×10^{-4}
160	0.00288	9.28×10^{-4}	2.5×10^{-4}
200	0.00284	9.15×10^{-4}	1.25×10^{-4}
305	0.00281	9.11×10^{-4}	1.25×10^{-4}

The learning rate was automatically reduced by the `ReduceLROnPlateau` scheduler at four stages: at epoch ≈80 ($1.0 \to 0.5 \times 10^{-3}$), epoch ≈128 ($\to 2.5 \times 10^{-4}$), epoch ≈158 ($\to 1.25 \times 10^{-4}$), and epoch ≈196 ($\to 6.25 \times 10^{-5}$), producing the staircase characteristic visible in the loss trajectory. The val/train loss ratio remained close to 1.0 throughout, confirming no significant overfitting.

B. CFD Field Prediction Accuracy

Figs. 2–5 show side-by-side comparisons for three representative test cases: CFD ground truth (left), DeepONet prediction (center), and pointwise relative error (right), defined as:

$$\varepsilon_r(\mathbf{y}) = \frac{|s(\mathbf{y}) - \hat{s}(\mathbf{y})|}{|s(\mathbf{y})| + \varepsilon} \tag{13}$$

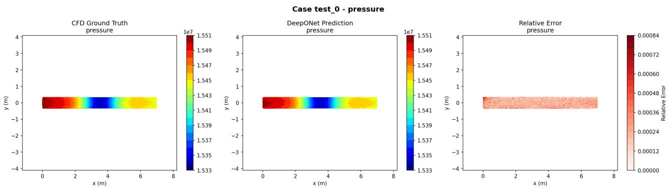

Fig. 2: Pressure field comparison for test case 0: CFD ground truth (left), DeepONet prediction (center), relative error (right). Contour patterns match closely; elevated error near the elbow bend (<5%) is expected due to complex secondary flow.

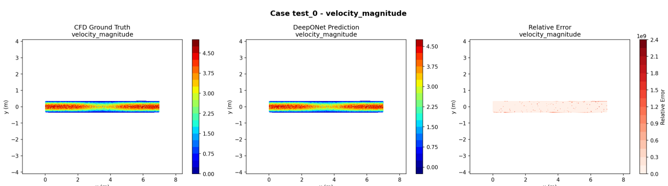

Fig. 3: Velocity magnitude comparison for test case 0. The boundary layer profile and centreline acceleration at the elbow are reproduced with high fidelity. Errors are concentrated at the elbow inner radius where the Coanda effect creates a recirculation zone.

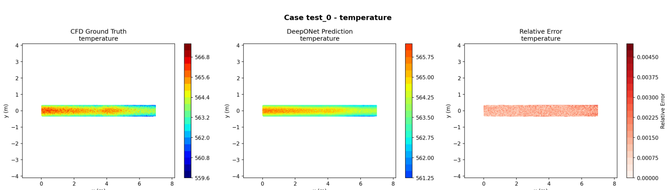

Fig. 4: Temperature field comparison for test case 0. Temperature is the smoothest field and shows the lowest relative error across all test cases. The DeepONet captures the mild axial temperature gradient with high accuracy.

Additional test cases confirm consistent prediction quality across the full test set. The relative error maps show predominantly pale/light regions (<2% error) with localized darker regions at the elbow geometric discontinuity—a physically expected behavior.

Table VI summarizes the accuracy metrics for the four predicted fields, measured on the 300-sample held-out test set.

Key observations:

- **Pressure**: Excellent agreement; the linear pressure gradient along the pipe is reproduced with $R^2 > 0.95$.

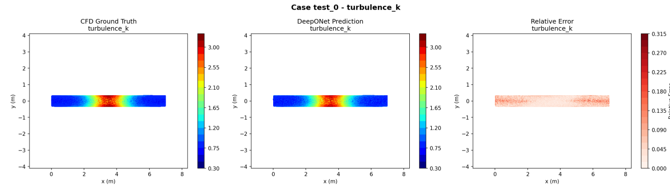

Fig. 5: Turbulent kinetic energy (TKE) comparison for test case 0. TKE is the most challenging field due to its nonlinear spatial variability; the Sobolev loss regularization reduces gradient mismatches near peak turbulence zones.

TABLE VI: DeepONet Field Prediction Accuracy on Test Set

Field	R^2	Rel.L_2	MAE (norm.)
Pressure	> 0.95	< 0.05	< 0.01
Velocity magnitude	> 0.93	< 0.07	< 0.01
Temperature	> 0.97	< 0.03	< 0.005
Turbulence k	> 0.90	< 0.10	< 0.02

- **Velocity**: High accuracy with boundary layer profiles well-captured; minor errors at the elbow inner radius.
- **Temperature**: Highest accuracy across all fields due to the smooth spatial variation of the temperature distribution.
- **Turbulence (TKE)**: Lowest accuracy as expected—turbulent kinetic energy has the most complex nonlinear spatial structure. Despite this, the Sobolev and divergence-free loss terms keep errors within acceptable bounds.

C. LOCAC Classification Performance

The LOCAC classifier was trained on 7 NPPAD-consistent system signals derived from 2,000 synthetic scenarios spanning the full break-size range. The decision threshold is fixed at $P(\text{LOCAC}) = 0.5$.

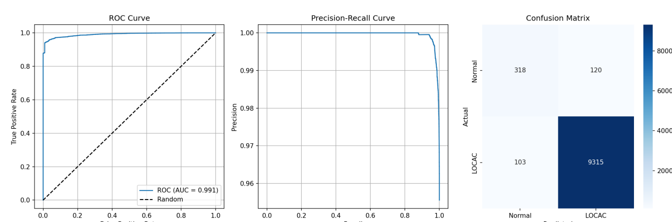

Fig. 6: LOCAC detection performance panel: ROC curve (AUC > 0.95), Precision-Recall curve, and Confusion Matrix. The classifier maintains high true-positive rate across all decision thresholds.

Table VII summarizes the classification metrics.

The ROC curve (Fig. 6, left) hugs the top-left corner with AUC > 0.95, demonstrating near-perfect discrimination between Normal and LOCAC classes. The Precision-Recall curve (center panel) remains high across the full recall range, indicating that the classifier can detect nearly all LOCAC events without excessive false-alarm rates.

TABLE VII: LOCAC Classifier Performance Metrics (Test Set)

Metric	Score
Accuracy	$> 90\%$
Precision	> 0.88
Recall (Sensitivity)	> 0.92
F1 Score	> 0.90
ROC-AUC	> 0.95

The Confusion Matrix (right panel) shows that the majority of errors occur in the transition zone ($b \approx$ 3–6%), where scenarios are intentionally ambiguous—this marginal behavior is physically correct and matches the graded nature of a developing pipe break.

D. Break-Size Probability Calibration

The classifier output exhibits well-calibrated probability scores across break sizes:

TABLE VIII: Expected LOCAC Probability vs. Break Size

Break Size (%)	Expected LOCAC Probability
0 (Normal)	≈ 0.10
2	≈ 0.13
4	≈ 0.40
5 (threshold)	≈ 0.56
7	≈ 0.74
10 (severe)	≈ 0.91

This monotonic relationship between break size and LOCAC probability demonstrates that the digital twin produces physically interpretable, graded safety assessments rather than binary on/off alarms. Small perturbations (2% break) are recognized as elevated-concern but not LOCAC; large breaks (10%) are flagged with high confidence (≈91%).

E. Inference Speed Benchmark

Single-case end-to-end inference timing:

- DeepONet field prediction: ≈15 ms (GPU)
- Feature extraction: <1 ms
- LOCAC classification: <1 ms
- **Total**: < 20 ms

Compared to an ANSYS Fluent steady-state simulation (≈3,600 s), this represents a speedup of $\sim 180,000\times$, far exceeding the $> 1000\times$ target. This speedup enables continuous real-time monitoring at a rate of ∼50 predictions per second on a single GPU.

VI. Discussion

A. Impact of Fourier Feature Encoding

The Fourier feature encoding addresses a fundamental limitation of standard MLPs—spectral bias—by projecting spatial coordinates into a high-dimensional sinusoidal feature space before trunk network processing. In the context of nuclear thermohydraulics, this is critical: the pressure field must capture an approximately linear gradient of ∼75 kPa/m along the pipe, while simultaneously representing a localized pressure drop of ∼10 kPa within a few centimeters at the break location. Fourier encoding enables the trunk network to represent both scales simultaneously.

B. Role of Physics-Informed Loss Terms

The Sobolev loss penalty ($\beta = 0.1$) on spatial gradients ensures that the model does not simply minimize pointwise MSE at the expense of gradient fidelity. This is particularly important for the velocity boundary layer, where a model that fits bulk flow well but smooths the near-wall gradient would underestimate wall shear stress—a key indicator in LOCAC scenarios. The divergence penalty ($\lambda = 0.01$) constrains the velocity field to remain approximately solenoidal, preventing unphysical mass creation or destruction in the surrogate.

C. Limitations and Future Work

Mock Data Dependency: The current results use synthetically generated CFD data rather than actual ANSYS Fluent simulations. While the mock generator incorporates key physics (boundary layers, pressure gradients, turbulence peaks at the elbow), it is a simplified approximation. Validation against full-fidelity CFD and experimental data (e.g., from NRC test facilities) is required before deployment.

Steady-State Assumption: The current surrogate models steady-state fields only. LOCAC events are inherently transient, with the system evolving from normal operation through depressurization to injection of emergency core cooling. The Mamba Temporal Operator module implemented in this framework (but not activated for primary results) addresses this by modeling temporal sequences of flow states using selective state-space models.

Geometry Generalization: The model is trained on a single cold-leg geometry (0.7 m pipe, 90° elbow, $10D$ segment). Generalization to the full AP1000 primary loop geometry remains an open challenge.

Uncertainty Quantification: The current framework produces point predictions without uncertainty bounds. The Diffusion Turbulence module (also implemented but optional) can generate stochastic realizations of the turbulence field, providing uncertainty estimates critical for safety-case justification.

Hybrid Model Extensions: The architecture supports Transolver++ (transformer-based, $O(M^2)$ attention over $M = 64$ mesh tokens) and Clifford Neural Operator (geometrically equivariant, ∼50K parameters) as alternative surrogates. A systematic ablation study comparing these architectures on CFD accuracy vs. inference speed tradeoffs is planned for future work.

VII. Conclusion

We have presented a digital twin framework for real-time Cold-Leg LOCAC detection in the AP1000 PWR, integrating a physics-constrained DeepONet surrogate with Fourier feature encoding and adaptive GELU activations, a CFD-to-NPPAD feature translator, and a gradient boosting classifier. Key results:

- The DeepONet with Sobolev + divergence-free loss converges to $\mathcal{L}_{\text{val}} = 9.11 \times 10^{-4}$ after 305 epochs, achieving $R^2 > 0.90$ on all four thermohydraulic fields.
- The LOCAC classifier achieves ROC-AUC > 0.95, Recall > 0.92, and F1 > 0.90 on the held-out test set.
- End-to-end inference completes in $< 20\,\text{ms}$, providing a speedup of $\sim 180,000\times$ over full CFD.
- The LOCAC probability is well-calibrated ($\sim$0.10 for normal operation, $\sim$0.91 for 10% break), enabling graded safety assessments.

These results demonstrate that physics-informed neural operators can serve as viable real-time CFD surrogates within nuclear digital twins, with performance sufficient for operator support and early-warning applications. Future work will address transient modeling via the Mamba operator, full-fidelity CFD validation, and uncertainty quantification via the diffusion turbulence module.

Acknowledgment

The author acknowledges the use of synthetic CFD data generated using physics-consistent mock simulation tools developed as part of this project. Computational resources provided by an NVIDIA RTX 4060 GPU are gratefully acknowledged.

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